

Evaluating the Reliability of Explainable AI in Ensemble-Based Intrusion Detection Systems

THESIS DEFENSE

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The research problem

An accurate detector can still produce unreliable explanations.

Repeated rankings may look consistent while masking tests show little predictive relevance.

A new traffic domain can change both detection and explanation behavior.

Research questions

RQ1 How repeatable are local explanations?

RQ2 How faithful are they under controlled masking tests?

RQ3 How do detection and explanations transfer?

RQ4 How much do independent explainers agree?

Research gap and contribution

Evaluate explanation reliability separately from detection accuracy.

Compare repeatability with a matched random masking control.

Test frozen detectors on a second dataset and include neural models.

Report LLM classification failures and explanation coverage.

Datasets and shared representation

NF-UNSW-NB15-v2: 1,986,745 cleaned flows.

NF-CSE-CIC-IDS2018-v2: 17,129,715 cleaned flows.

39 shared numeric features after removing identifiers and targets.

Binary transfer uses compatible labels. Multiclass evaluation stays within each dataset.

Evaluation protocol

Group identical feature vectors before the 70/15/15 split.

Cap training at 200,000 flows, validation at 40,000 and test at 80,000.

Repeat detector fitting with seeds 7, 42 and 1337.

Freeze preprocessing and the source model for target evaluation.

Model comparison

Tree models: decision tree, random forest and XGBoost.

Ensemble: soft voting over the three tree models.

Neural models: shallow MLP, deep MLP and feature-axis CNN.

84 fitted configurations and 126 evaluation cells.

Explanation evaluation

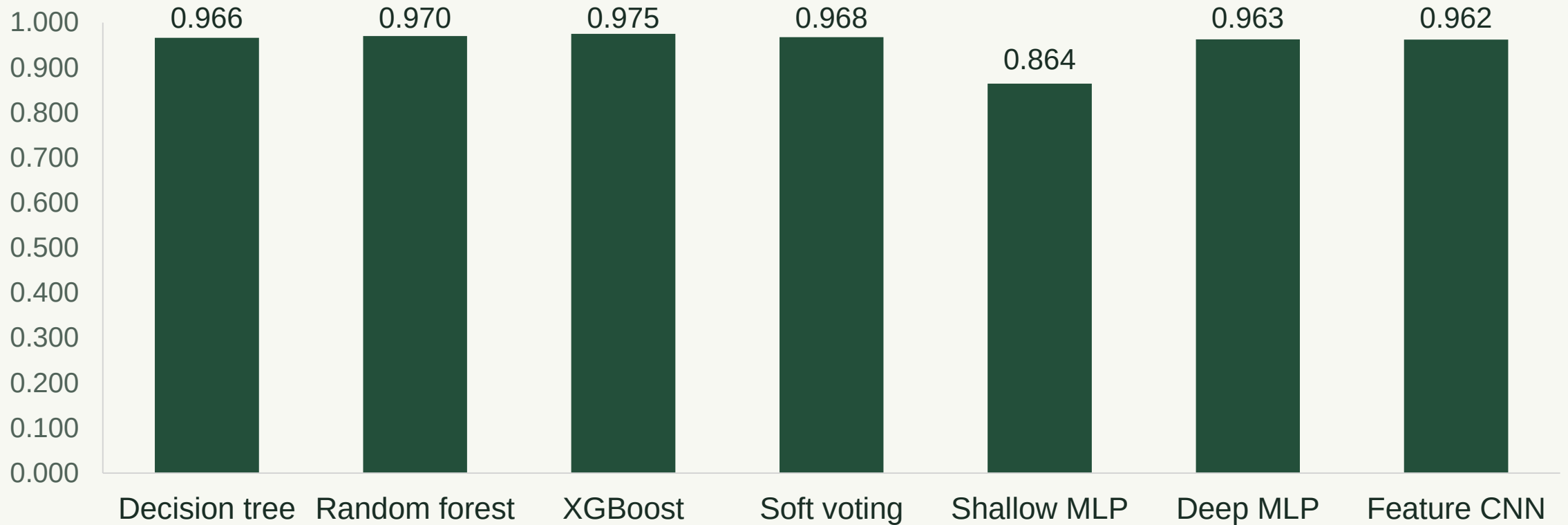
LIME: 20 balanced unique feature groups per domain.

Three explainer seeds and 5,000 perturbations per explanation.

Jaccard@5 measures overlap between repeated feature sets.

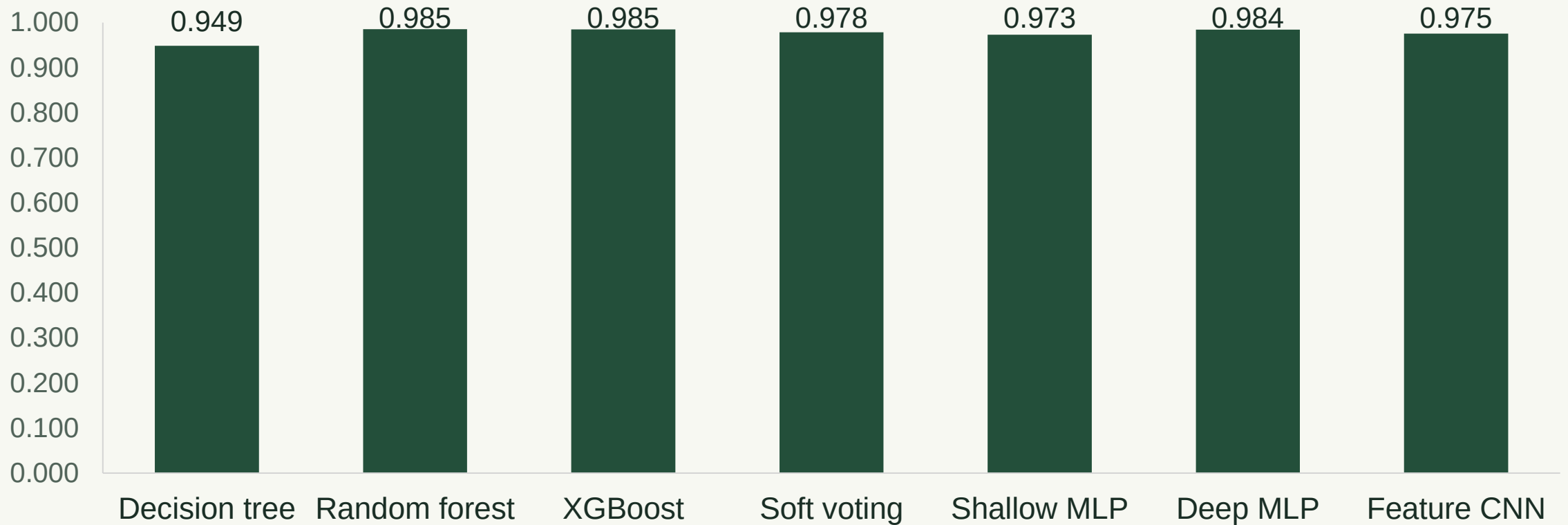
Masking advantage compares selected features with random masks.

Binary detection on UNSW



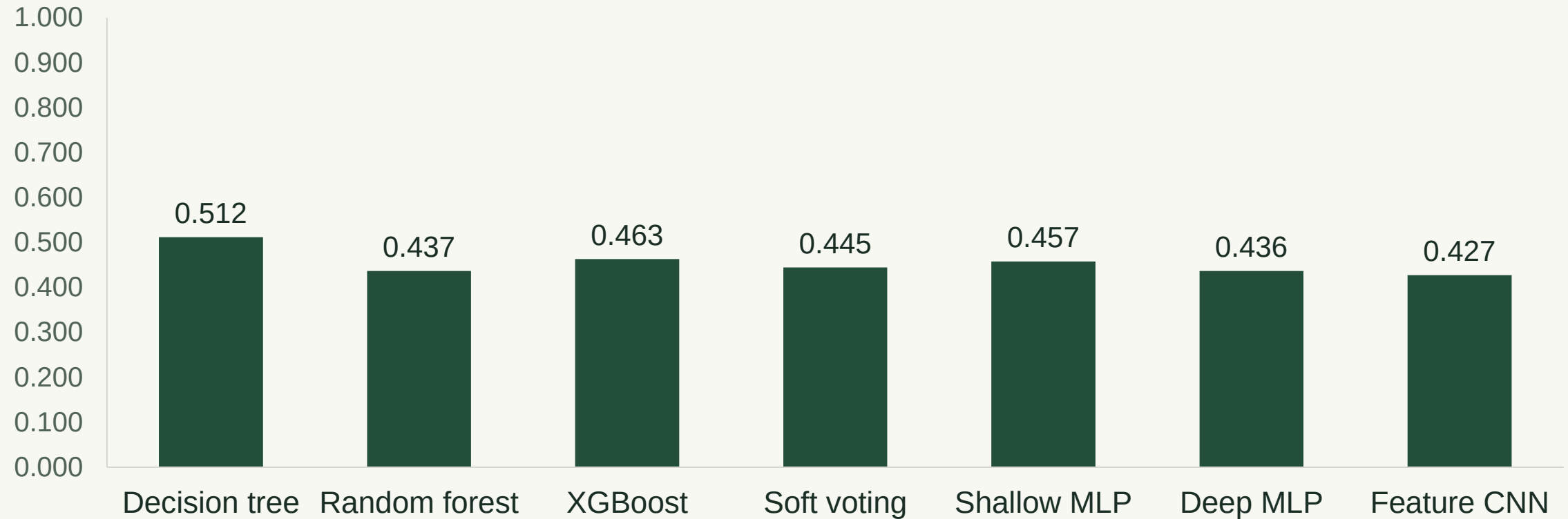
Macro-F1 means across three seeds. XGBoost leads at 0.9750.

Binary detection on IDS2018



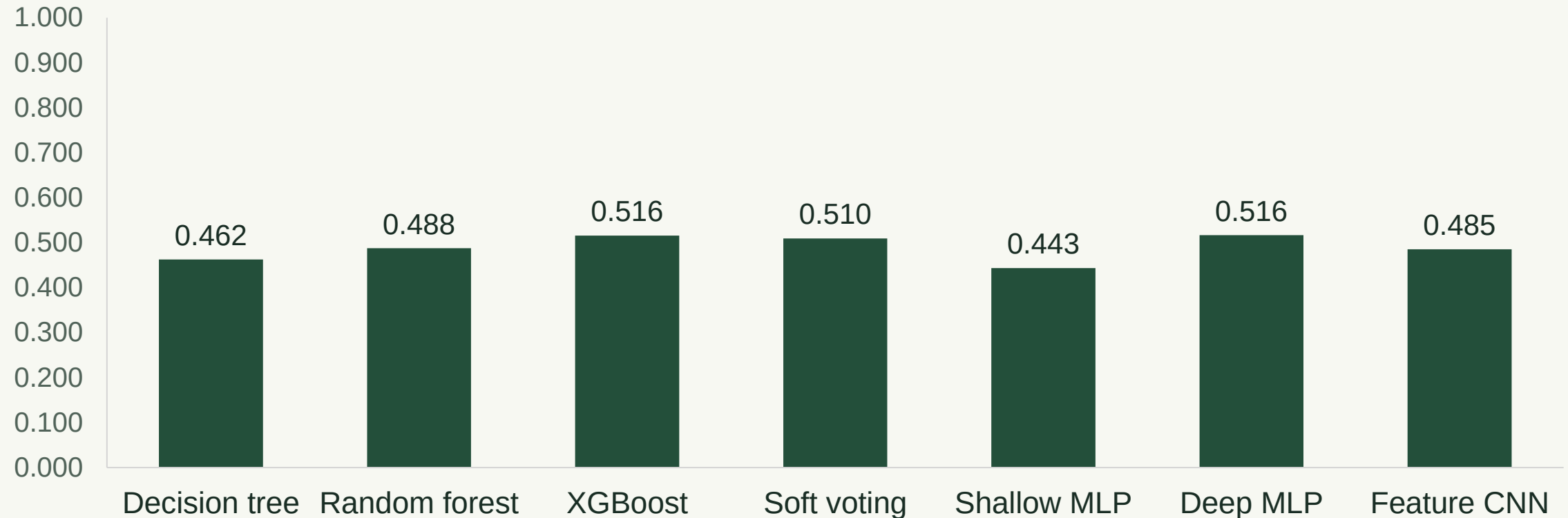
Macro-F1 means across three seeds. Random forest leads at 0.9852.

Transfer from UNSW to IDS2018



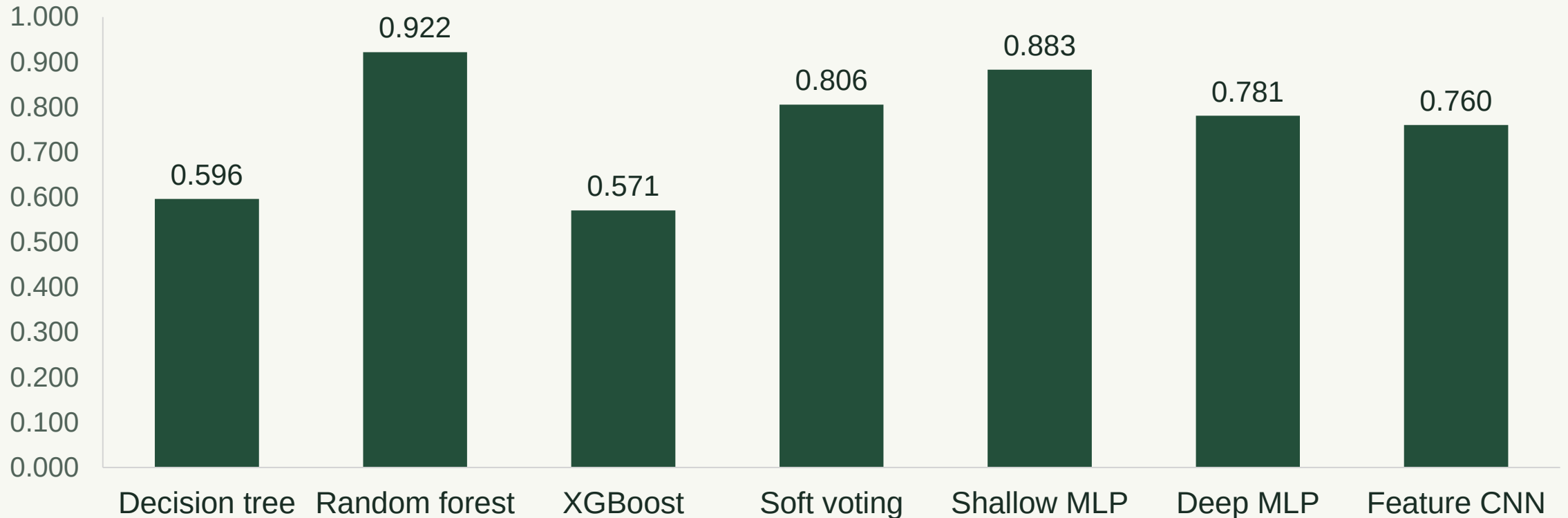
Frozen binary models. All seven families deteriorate on the target domain.

Transfer from IDS2018 to UNSW



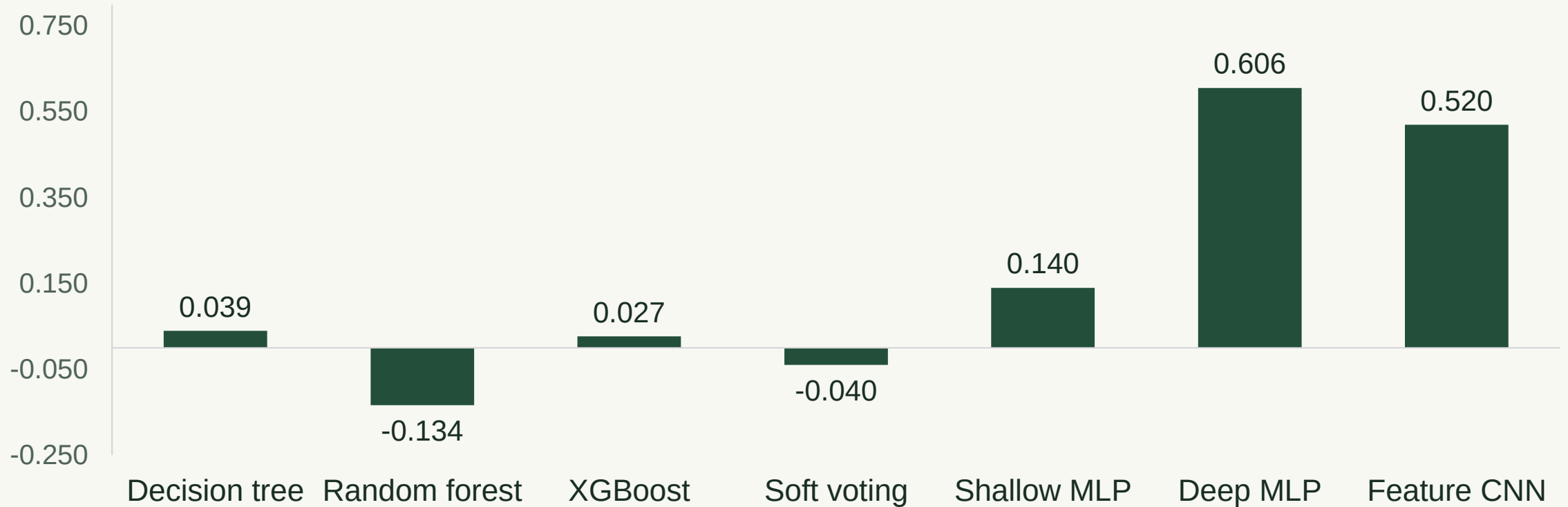
Macro-F1 means across three seeds. The best target mean remains about 0.516.

Explanation stability after transfer



UNSW-trained models on IDS2018. Random forest Jaccard@5 = 0.9222.

Masking advantage after transfer



Random forest: -0.1338, 95% interval [-0.2034, -0.0476].

Stability does not establish faithfulness

Random forest remains highly repeatable after transfer.

Its median-mask advantage becomes negative on IDS2018.

Signed ranking and mean masking can change the result.

Interpret the score together with its perturbation protocol.

Historical result audit

483 matched cases per tree model support reconstruction.

Within-model stability–advantage correlations are positive.

The arithmetic reconstruction found no material reversal.

The old random split has 88.97% exact-input overlap.

RQ4: agreement between explainers

Historical LIME and TreeSHAP rankings show partial agreement.

Random forest rank correlation is 0.676 in that protocol.

The same comparison shares only two of the top five features.

Rank correlation and top-feature overlap answer different questions.

Language-model extension

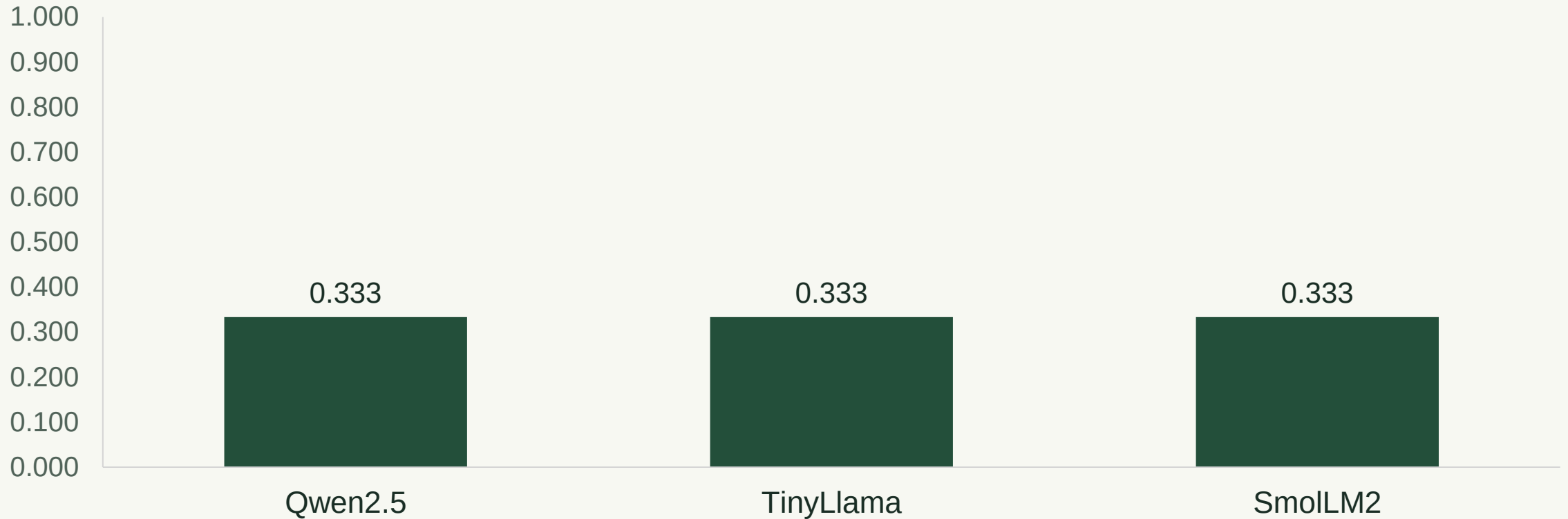
Qwen2.5-0.5B, TinyLlama-1.1B and SmolLM2-1.7B.

400 classification cases per model under the frozen prompt.

80 own-decision and 80 detector-grounded explanations per model.

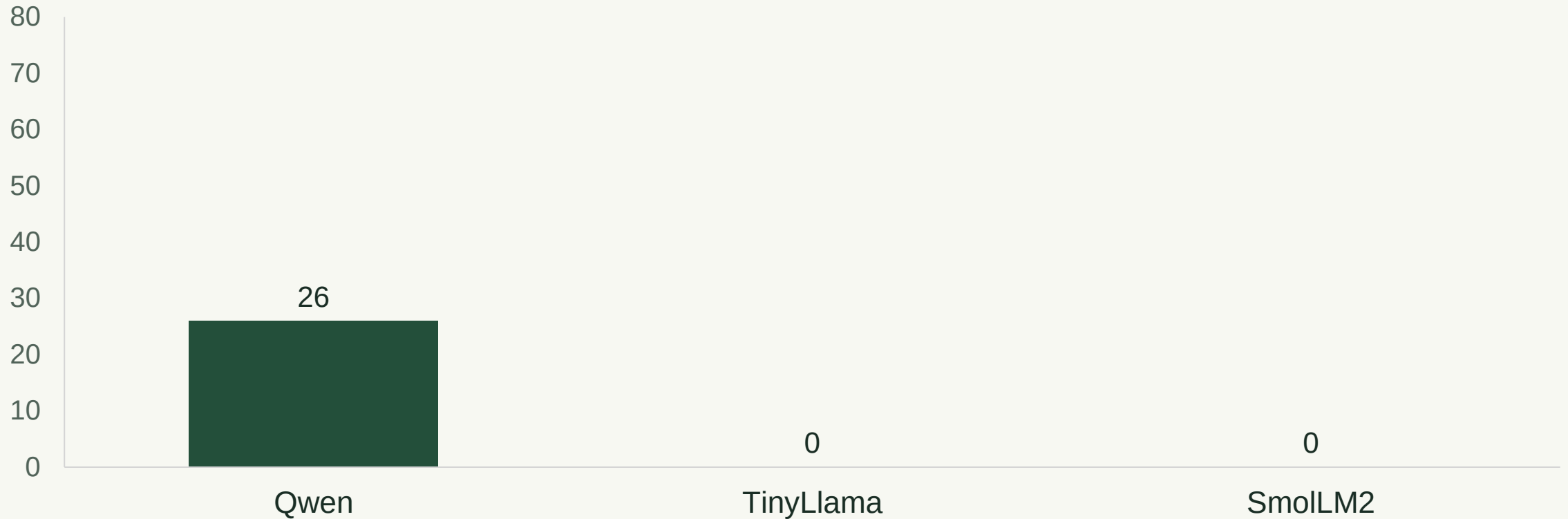
1,200 classifications and 240 matched explanation pairs.

LLM classification under the frozen prompt



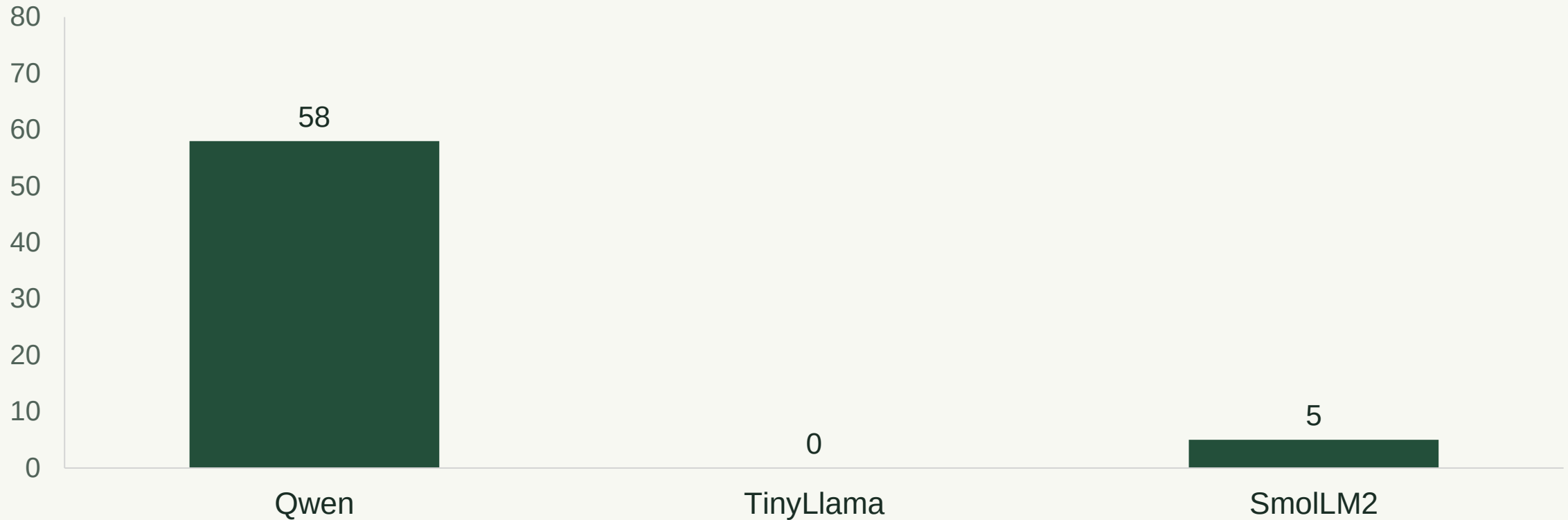
Each model predicts one class throughout its balanced cohort. Balanced accuracy = 0.5000.

Valid own-decision explanations



Valid feature-list outputs out of 80 per model. Masking scores apply only to valid outputs.

Valid detector-grounded explanations



Valid feature-list outputs out of 80 per model. Masking scores apply only to valid outputs.

Conclusions

High within-dataset accuracy does not guarantee transfer.

Stable explanations can have weak masking advantage.

The small LLMs fail classification under the tested protocol.

The audit supports narrower, protocol-specific conclusions.

Limitations and future work

Use chronological splits and evaluate on operational traffic.

Repeat explanation tests across more cases and training seeds.

Test conditional perturbations and analyst usefulness.

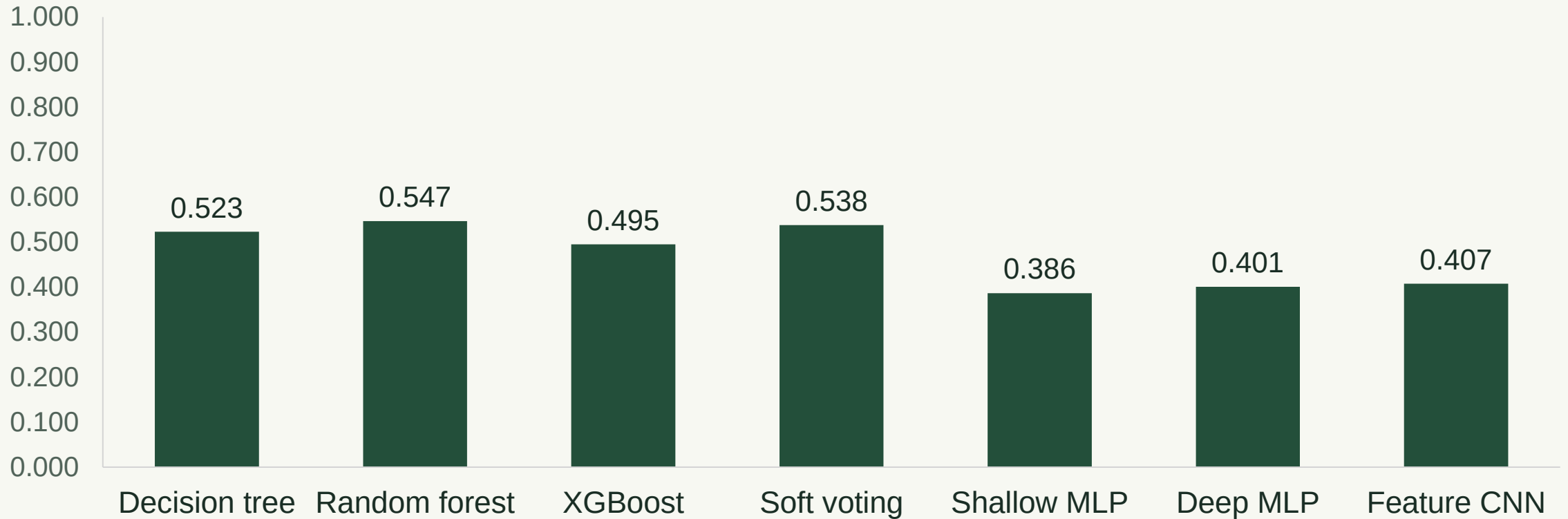
Evaluate larger LLMs with controlled prompt and scoring ablations.

Questions

How should explanation reliability change an IDS decision?

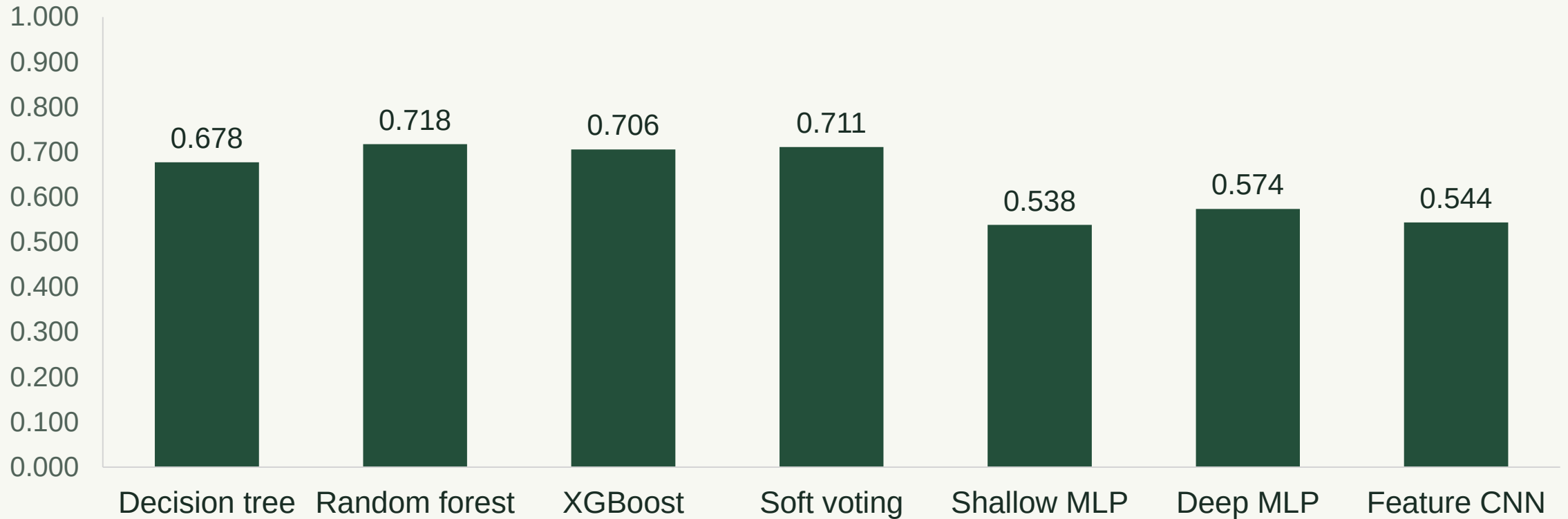
Backup slides contain multiclass results and protocol details.

Backup: UNSW multiclass detection



Native attack taxonomy. Macro-F1 means across three seeds.

Backup: IDS2018 multiclass detection



Native attack taxonomy. Cross-dataset multiclass labels are not aligned.

Backup: neural training protocol

Shallow MLP: one hidden layer with 64 units.

Deep MLP: 256, 128 and 64 units with 0.1 dropout.

Feature CNN: 16 and 32 channels, kernel size 3, dense layer 64.

AdamW, 20-epoch cap, patience 4, batch size 1,024.

Backup: sources and evidence

Original thesis and preserved expanded report.

Detection: results/study/across_seed_summary.csv

Explanations: results/study/xai/aggregate.csv

LLM coverage: results/study/llm/masking_coverage.csv